

JavaScript Concepts & Examples

1. String Methods

JavaScript provides various methods to manipulate and search strings.

Concepts:

- The `.search()` method is used to find the position of a substring in a string.

Example:

javascript



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```
let x = "Hello world"; console.log(x.search("world")); // Output: 6
```

2. Handling Errors with Try-Catch

JavaScript uses `try...catch` blocks to handle errors gracefully.

Concepts:

- `try` block contains the code that may throw an error.
- `catch` block executes when an error occurs.

Example:

javascript



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```
try { alert("hello"); // Typo in original code: should be `alert`, not `allert` } catch  
(error) { console.log(error); // Logs the error to the console }
```

3. DOM Manipulation: Changing Text Dynamically

JavaScript allows us to modify HTML elements dynamically using `innerHTML`.

Concepts:

- `document.getElementById("id").innerHTML` is used to get or modify an element's content.

Example:

javascript



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```
function myFunction() { let textele = document.getElementById("demo"); textele.innerHTML = textele.innerHTML.replace("moodbidri", "accelerlab"); }
```

When the button is clicked, "moodbidri" is replaced with "accelerlab" inside the `<p>` tag.

4. Arrow Functions

Arrow functions provide a shorter syntax for writing functions.

Concepts:

- Arrow functions use `=>` instead of `function` keyword.
- They are useful for short, single-expression functions.

Example:

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```
const myFunction2 = () => { console.log("I'm an arrow function"); } myFunction2();
```

5. Array Mapping (`map()`)

The `.map()` function is used to apply a function to each element in an array.

Concepts:

- Takes a function as an argument and applies it to each element.

- Returns a new array with modified values.

Example:

javascript



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```
let a = [10, 20, 30, 40]; let result = a.map((item) => console.log(item * item));
```

This prints the square of each element.

6. Destructuring Arrays

Destructuring allows extracting values from arrays easily.

Concepts:

- Uses `[ ]` to assign array elements to variables.

Example:

javascript



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```
let a = [1, 2, 3, 4]; const [first, second, third, fourth] = a; console.log(first); //  
Output: 1
```

7. Object Destructuring

Extracts properties from objects into variables.

Concepts:

- Uses `{ }` to assign object properties to variables.

Example:

javascript



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```
const person = { name: 'John', age: 35, job: 'developer' }; const { name, age, job } = person; console.log(name, age, job); // Output: John 35 developer
```

8. Spread Operator (...)

The spread operator is used to merge or copy arrays/objects.

Concepts:

- [...array1, ...array2] merges two arrays.
- {...object1, ...object2} merges objects.

Example: Array Spread

javascript



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```
let d = [1, 2, 3, 4]; let b = [5, 6, 7, 8]; let c = [...d, ...b]; console.log(c); // Output: [1,2,3,4,5,6,7,8]
```

Example: Object Spread

javascript



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```
const myVehicle = { brand: 'Ford', model: 'Mustang', color: 'red' }; const updateMyVehicle = { type: 'car', year: 2021, color: 'yellow' // Overwrites color property }; const myUpdatedVehicle = { ...myVehicle, ...updateMyVehicle }; console.log(myUpdatedVehicle); // Output: { brand: 'Ford', model: 'Mustang', color: 'yellow', type: 'car', year: 2021 }
```

Note: The `color` property is overwritten by the new value.

9. Ternary Operator (? :)

A shorthand for if-else statements.

Concepts:

- `condition ? expression_if_true : expression_if_false`

Example:

javascript



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```
let x = 10; x === 10 ? console.log("Match") : console.log("Not Match"); // Output: Match
```

If `x` is `10`, "Match" is logged; otherwise, "Not Match" is logged.